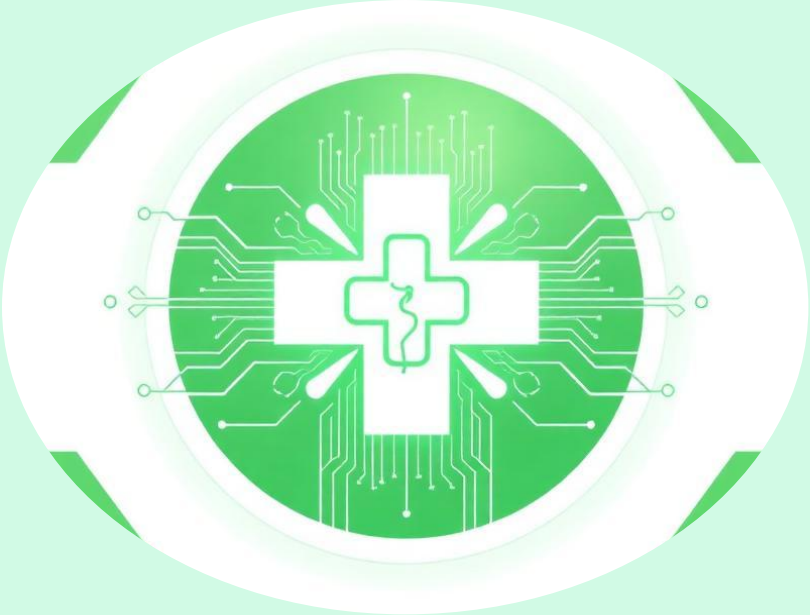
WellBot

AI-Powered Healthcare Chatbot

Your AI Wellness Partner, Always By Your Side



Presented by: Team-D

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March 2026

Meet Team-D

M. Geethika

Backend Developer & AI Integration

Rashi Rathore

Frontend Developer & UI/UX Designer

R. Jansi Raju, Sana Afrin

Database & Language switching in chatbot

Technical Stack

Frontend: HTML, CSS, React.js

Backend: Python, Flask API

AI/ML: Groq AI, RandomForest ML

Database: SQLite DB

Project Overview

WellBot is an AI-powered healthcare chatbot that provides personalized wellness guidance across multiple domains including mental health, fitness, nutrition, motivation, and stress relief.

Key Highlights:

- 6 Specialized AI Chat Specialists for different wellness areas
- AI Wellness Predictor powered by RandomForest ML model
- BMI Calculator for health metrics tracking
- Multi-language support (5+ languages) for global accessibility
- Real-time mood prediction and productivity scoring
- Interactive dashboard with weekly activity charts and wellness trends
- Chat export functionality for record keeping

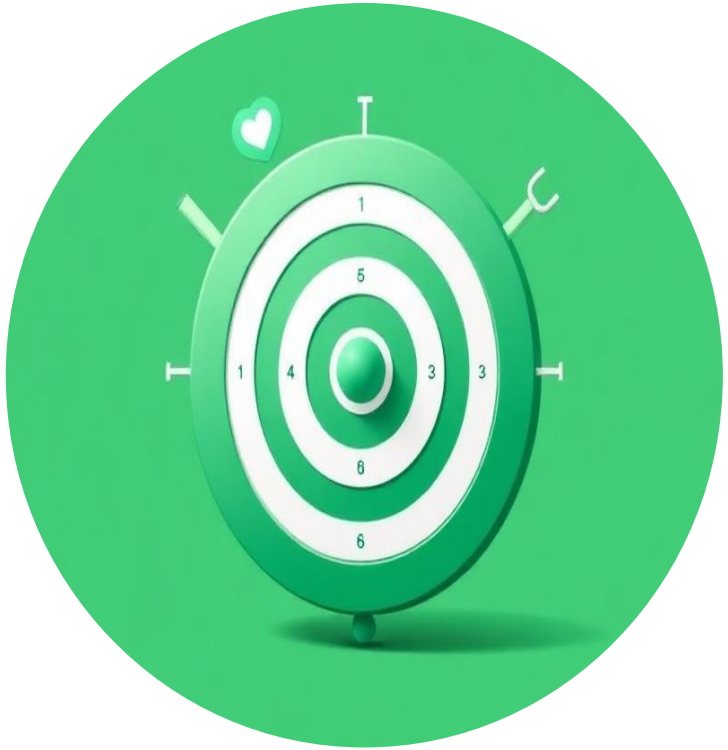
Vision & Mission

🎯 Our Vision

To revolutionize personal healthcare by making AI-driven wellness guidance accessible, affordable, and available 24/7 to everyone — breaking barriers of language, location, and cost.

🚀 Our Mission

- Provide expert-level wellness guidance through specialized AI chatbots
- Enable early wellness prediction using machine learning models
- Support multilingual conversations for inclusive healthcare
- Empower users to track and improve their daily wellness habits
- Deliver a seamless, user-friendly healthcare experience



Proposed Solution

🧠 AI Chat Specialists

6 domain-specific chatbots — Mental Health, Fitness Coach, Nutrition, Motivation, Stress Relief, and General Chat — each powered by Groq AI with tailored system prompts.

📊 ML Wellness Predictor

RandomForest ML model that predicts mood, stress level, and productivity score based on daily inputs like sleep, exercise, caffeine, screen time, and more.

🌐 Multi-Language Support

14+ language support including Hindi, Telugu, Tamil, and more — AI responds in user's preferred language for inclusive healthcare access.

📈 Smart Dashboard

Interactive dashboard with weekly activity charts, mood trends, streak tracking, and feedback ratings — giving users a holistic wellness overview.

Our Development Process

01

Research & Planning

Identified key wellness domains, studied existing healthcare chatbots, and defined project scope and requirements.

02

UI/UX Design

Designed dark-themed, modern UI with intuitive navigation, dashboard layout, and responsive chat interfaces.

03

Frontend Development

Built with HTML, CSS & React.js — creating reusable components, routing, and a seamless user experience.

04

Backend & AI Integration

Developed Flask API endpoints, integrated Groq AI for chat, trained RandomForest ML model for wellness prediction.

05

Testing & Deployment

Comprehensive testing across browsers, multilingual testing, performance optimization, and final deployment.

Application Workflow

1. User visits Landing Page
2. Sign Up / Login with credentials
3. Access Dashboard with wellness overview
4. Choose a Chat Specialist (Mental Health, Fitness, etc.)
5. Chat with AI in preferred language
6. Use AI Wellness Predictor (enter daily stats)
7. Get mood, stress & productivity predictions
8. Use BMI Calculator for health assessment
9. Track progress via charts & trends
10. Export chat history for records

System Architecture

Frontend Layer

HTML, CSS, React.js
Responsive UI, Dark Theme, Multi-language

API Layer (Flask)

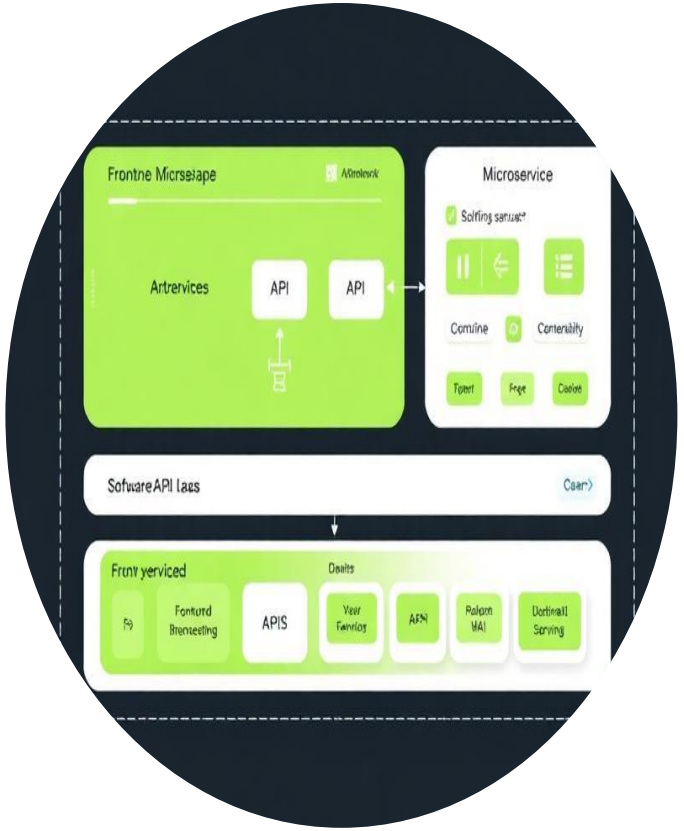
RESTful API Endpoints
Authentication, Chat, Prediction, BMI

AI/ML Layer

Groq AI (Chat Engine)
RandomForest ML (Wellness Predictor)

Database Layer

SQLite DB
Users, Chats, Feedback, Wellness Data



Technical Stack

Frontend

- HTML5 & CSS3 — Structure and styling
- React.js — Component-based UI framework
- Responsive Design — Mobile & desktop compatible

Backend

- Python — Core programming language
- Flask — Lightweight web framework & REST API
- Flask-CORS — Cross-origin resource sharing

AI & ML

- Groq AI — Fast LLM inference for chat responses
- RandomForest ML — Wellness prediction model
- Scikit-learn — ML model training and evaluation

Database

- SQLite — Lightweight relational database
- User auth, chat history, feedback storage

BMI Calculator Feature

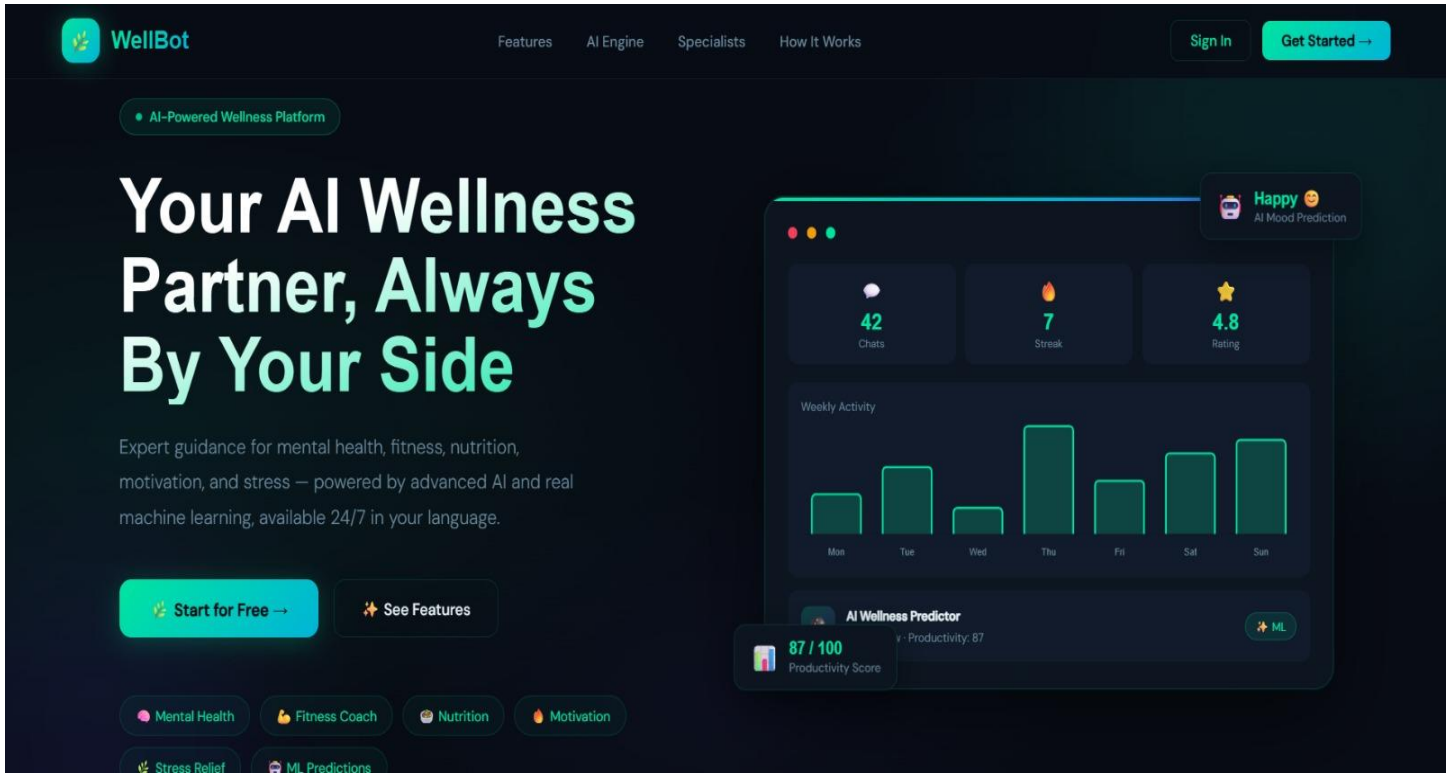
WellBot includes a built-in BMI (Body Mass Index) Calculator that helps users quickly assess their body weight category based on height and weight inputs.

How It Works:

1. User enters their height (cm) and weight (kg)
2. BMI is calculated using the standard formula: weight / (height^2)
3. Result is categorized: Underweight, Normal, Overweight, or Obese
4. Personalized health tips are provided based on the BMI category
5. Users can track BMI changes over time through the dashboard

💡 *BMI is a quick health indicator that complements the AI Wellness Predictor for comprehensive health assessment.*

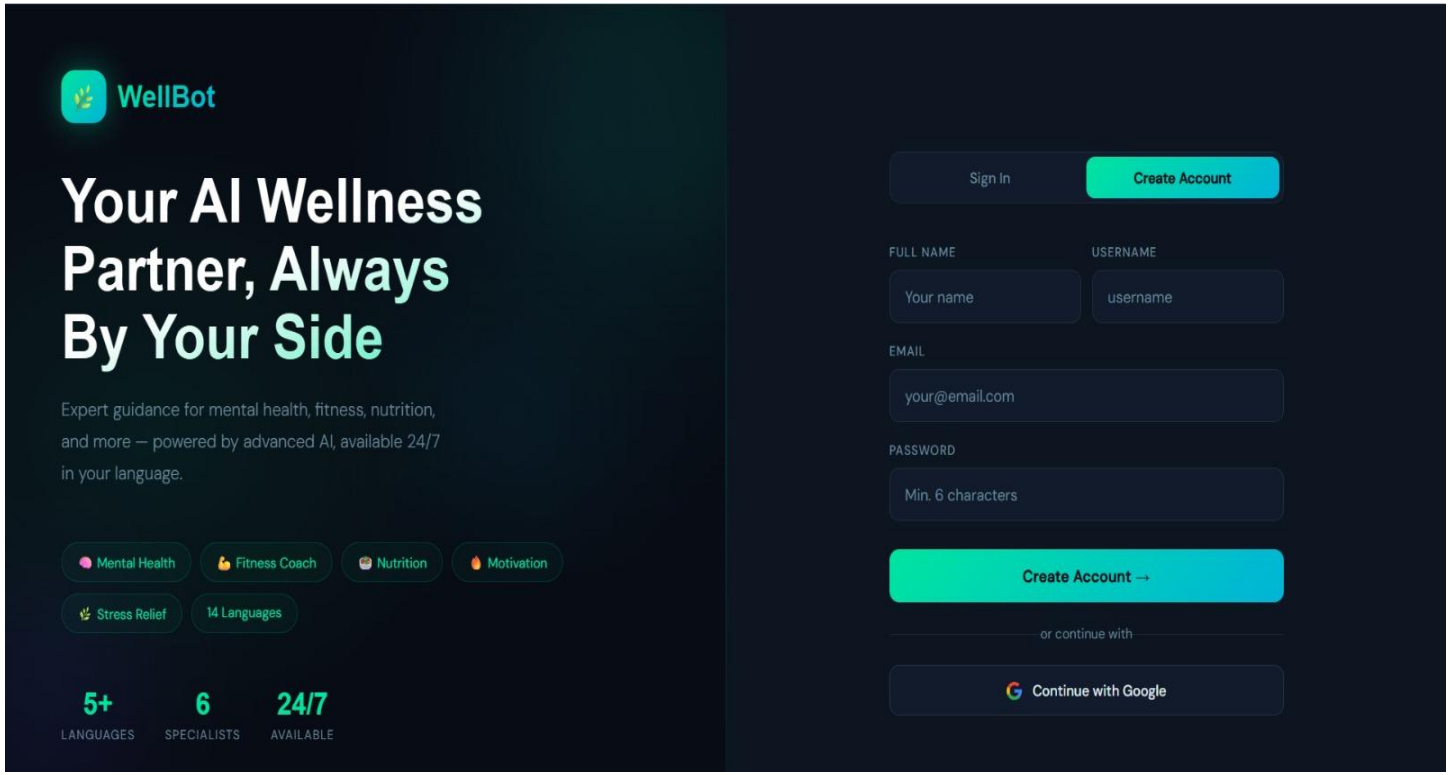
Landing Page



Key Elements:

- Modern dark-themed design with green accents
- Hero section with clear value proposition
- Interactive dashboard preview showing stats
- Category badges for wellness domains
- Call-to-action buttons: 'Start for Free' & 'See Features'
- Navigation with Features, AI Engine, Specialists sections

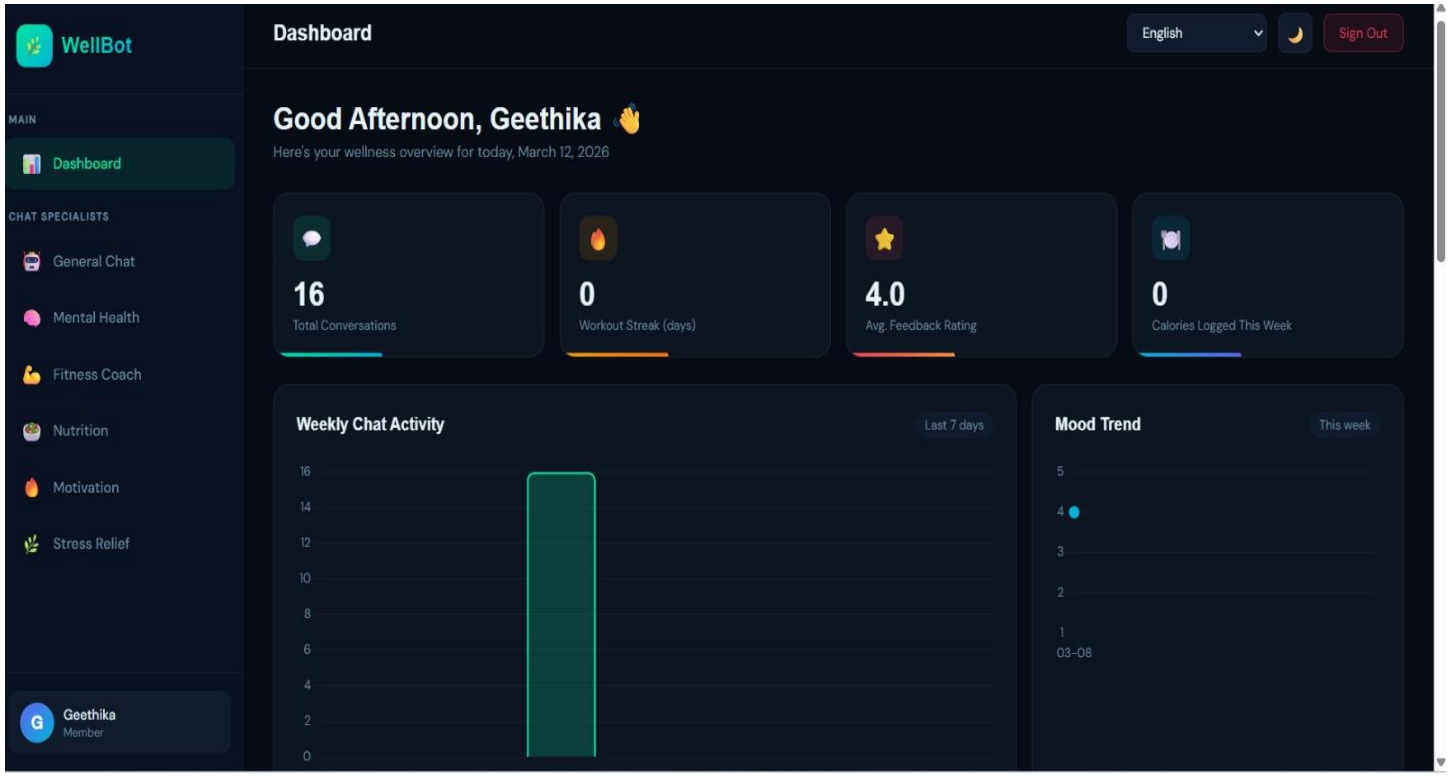
Login & Registration



Key Elements:

- Sign In / Create Account toggle tabs
- Full Name, Username, Email & Password fields
- 'Continue with Google' social login option
- Gradient CTA button for account creation
- Left panel shows platform features & stats
- Stats: 5+ Languages, 6 Specialists, 24/7 Available

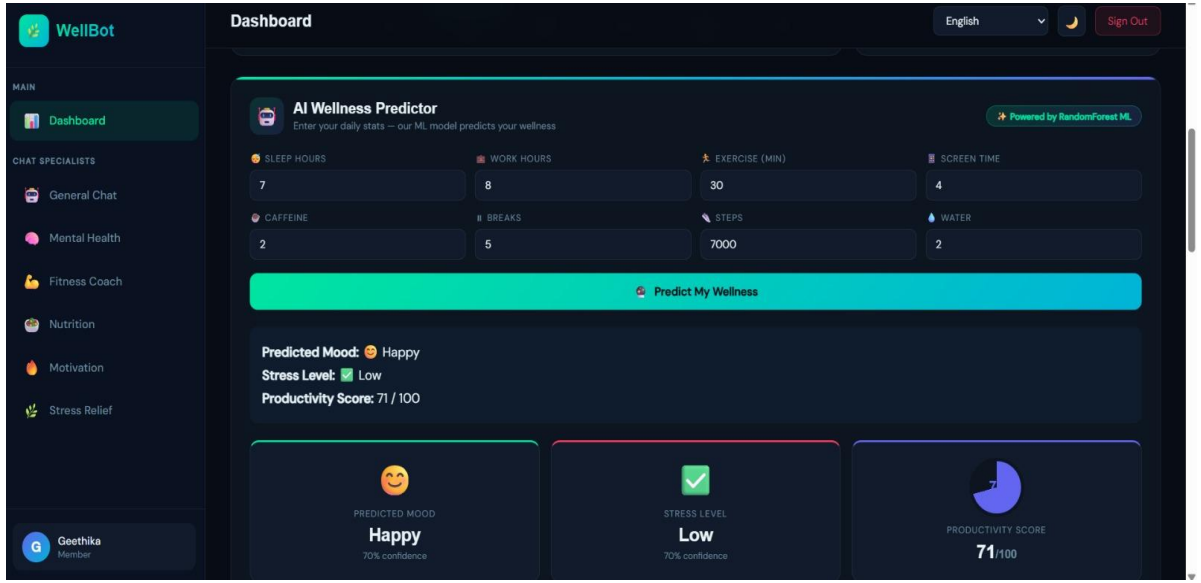
Dashboard & Analytics

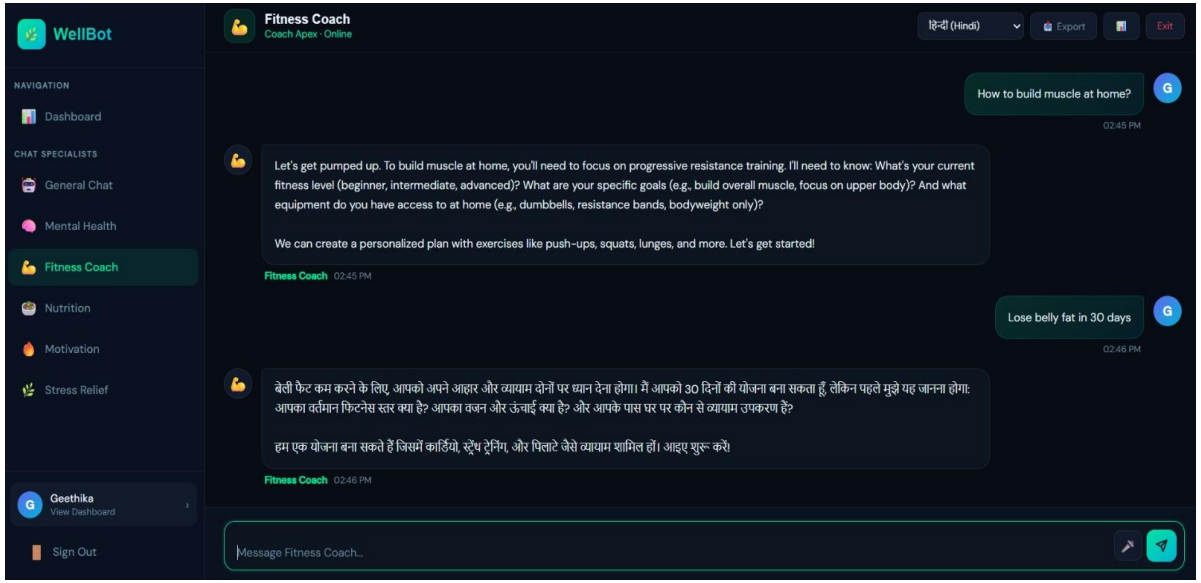


Key Elements:

- Personalized greeting with time-based message
- Stat cards: Conversations, Streak, Rating, Calories
- Weekly Chat Activity bar chart
- Mood Trend visualization over time
- Left sidebar with specialist navigation
- Language selector & dark mode toggle

AI Wellness Predictor & Chatbot





AI Wellness Predictor

- Enter daily stats: sleep, work, exercise, caffeine, steps, water, screen time, breaks
- ML model predicts Mood, Stress Level & Productivity Score
- Visual result cards with confidence percentages
- Powered by RandomForest ML algorithm

AI Chat Specialists

- 6 specialist chatbots for different wellness areas
- Real-time AI responses powered by Groq
- Multi-language support — responds in user's language
- Chat export and history tracking features

Feature Enhancements

🔔 Smart Notifications

Push notifications for hydration reminders, exercise breaks, and mood check-ins.

🤝 Community Forum

Peer support groups and wellness communities for shared motivation and experiences.

📊 Advanced Analytics

Monthly wellness reports, trend analysis, and personalized improvement suggestions.

⌚ Wearable Integration

Sync with fitness trackers and smartwatches for automatic health data tracking.

📱 Mobile App (PWA)

Progressive Web App support for native mobile-like experience on smartphones.

🩺 Doctor Integration

Connect with real healthcare professionals for advanced medical consultations.

🎮 Gamification

Achievement badges, wellness challenges, leaderboards, and reward points system.

🔐 Enhanced Security

Two-factor authentication, encrypted chats, and HIPAA-compliant data handling.

Thank You!

WellBot — Your AI Wellness Partner, Always By Your Side

WellBot demonstrates the power of combining AI, Machine Learning, and modern web technologies to create a comprehensive healthcare assistant. With 6 specialized chatbots, ML-powered wellness predictions, BMI calculator, and multi-language support — WellBot makes personalized healthcare guidance accessible to everyone, anytime, anywhere.

Team-D

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Questions? We'd love to hear from you! 🙌